James Chen

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Education

Columbia University — New York, NY
M.S. in Computer Science — *Expected Dec. 2027*
Dean's MS Academic Excellence Fellowship - $15,000

University of Massachusetts Lowell — Lowell, MA
B.S./M.S. in Computer Science; GPA: 4.0/4.0 — *Jan. 2024 – Dec. 2025*
Chancellor's Medals for Distinguished Academic Achievement

Skills

- **Programming Languages**: Java (4 yrs), Javascript/Typescript (4 yrs), Python (3 yrs), Go, C++, Kotlin, PHP, Lua, Rust, Bash, SQL, HTML/CSS
- **Frameworks and Libraries**: React, Spring Boot, Node.js, Vite, MUI, NestJS, Drizzle ORM, JUnit, Numpy, Pandas, Matplotlib, Scikit-learn, PyTorch, Typer
- **Databases and APIs**: PostgreSQL, MySQL, MongoDB, Redis, HTTPS, RESTful, GraphQL, WebSockets, Swagger
- **Tools and Platforms**: Git, GitHub, Docker, Nginx, systemd, AWS EC2/Lightsail, POSIX, npm/pnpm/bun, Neovim, Postman
- **AI**: OpenAI API, Kaggle, NLP, Neural Networks, Transformer/GPT, LoRA, RLHF, PPO

Work Experience

Software Engineering Intern — May. 2025 – Dec. 2025
UMass Lowell - Miner School of Computer and Information Sciences — *Lowell, MA*
- Migrated 28k+ AEPW (Alliance to End Plastic Waste) records into a analytical PostgreSQL schema, cutting query times by about 70% and improving data integrity.
- Designed and implemented AUDA, a Python analytics platform for data cleaning, preprocessing, model execution, and visualization.
- Built and evaluated machine learning workflows for sparse country-level time series, including Isolation Forest anomaly detection, Random Forest feature importance, and forecasting/reconstruction with automated hyperparameter optimization.
- Developed a Typer-based CLI to automate analytics workflows, improving reproducibility, maintainability, and developer productivity.

Backend Developer Intern — Mar. 2021 – Sep. 2021
Take 2 Health — *Shenzhen, China*
- Engineered the backend for a blood specimen management platform using Midway.js, implementing JWT-based authentication and RESTful APIs for user and specimen operations.
- Created Swagger API documentation to streamline integration and testing, and collaborated with frontend engineers to ensure reliable interface behavior.
- Architected a microservice-based system using gRPC for high-performance inter-service communication, improving scalability and modularity.

Backend Developer — Nov. 2018 – Feb. 2020
Juyingtong — *Zhuhai, China*
- Developed the backend for a WeChat mini-program for conference management using ThinkPHP 5 and MySQL, with Redis-backed session authentication and RESTful APIs for user and event operations.
- Built a location-aware attendee check-in system using QR code scanning, encrypted conference metadata, WeChat geolocation validation, and real-time attendance recording.
- Integrated the WeChat Pay API to support secure, real-time payment processing for conference registration and related services.
- Deployed the application on Alibaba Cloud Linux servers with Nginx reverse proxy and load balancing, using systemd for process management.

Projects

- **Bloxtopia**: A startup project targeting Roblox players. Users can complete tasks on the offerwall to earn Robux, which they can then redeem to their Roblox accounts. Next.js and Nodemailer were used in this project.
- **Nemotron**: An online challenge where competitors fine-tune a foundation LLM, NVIDIA Nemotron, using LoRA adapters. PEFT and vLLM were used in this project. Our team achieved an accuracy of 0.85.
- **(Snowy) Outpost**: A tool that creates a temporary high-performance DigitalOcean droplet (a cloud instance) and hosts a Minecraft server there. It automatically uploads the game directory on start and downloads it on stop.
- **Burrow**: An extensible command-line application platform that lets users create independent CLI applications (chambers) composed of extensions, which can start in under 20 milliseconds. There are built-in modules for chamber management, configuration editing, HTTP server startup, and self-build/update workflows.